POLYNOMIAL CONFIGURATIONS AND POINTWISE AVERAGES ALONG PIATETSKI–SHAPIRO PRIMES

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Abstract. For every $c > 1$ sufficiently close to 1, we prove that there is $\kappa > 0$ such that every sufficiently large $A \subset [N]$ of density at least $(\log N)^{-\kappa}$ contains

$$x,\ x + (p-1),\ x + (p-1)^2, \dots,\ x + (p-1)^k, \qquad p = \lfloor n^c \rfloor \in \mathbb{P}.$$

We also prove pointwise almost-everywhere convergence of the corresponding multiple ergodic averages.

1. Introduction and transference

The polynomial Szemerédi theorem of Bergelson and Leibman [3] generalizes Sárközy's square-difference theorem [19] to polynomial configurations in sets of positive upper density. Wooley and Ziegler restricted the common parameter to shifted primes [25]. Building on these results, Krause, the author, Tao and Teräväinen obtained the quantitative density bound $(\log N)^{-\kappa}$ for distinct-degree polynomial families [9, Theorem 1.2].

Szemerédi- and Green–Tao-type problems for arithmetically defined subsets of the primes have attracted particular interest when these sets have power-saving zero relative density. Examples include Piatetski–Shapiro primes [13, 11, 21] and primes with restricted digits, studied by Maynard [12], for which Burgin obtained related recurrence results [5].

For $c > 1$, put

$$\mathcal{S}_c := \{\lfloor n^c \rfloor : n \in \mathbb{N}\}, \qquad \mathbb{P}_c := \mathcal{S}_c \cap \mathbb{P}.$$

When c is sufficiently close to 1,

$$\frac{|\mathbb{P}_c \cap [N]|}{|\mathbb{P} \cap [N]|} \sim N^{-(1-1/c)}.$$

This follows from the Piatetski–Shapiro prime number theorem [18], which gives $|\mathbb{P}_c \cap [N]| \sim N^{1/c}/\log N$, together with the ordinary prime number theorem $|\mathbb{P} \cap [N]| \sim N/\log N$. Thus $\mathbb{P}_c$ has zero relative density among the ordinary primes, with a power saving. Our first theorem shows that, for distinct-degree polynomial families, restricting the gap size parameter to $\mathbb{P}_c - 1$ still permits a density threshold $(\log N)^{-\kappa}$ for the integer set containing the configuration. Up to a change in $\kappa > 0$, this matches the form of Shao and Wang's bound for unrestricted integer parameters [20, Theorem 1.1] and the bound of Krause, the author, Tao and Teräväinen for shifted-prime parameters [9, Theorem 1.2].

Fix throughout

$$P_0 = 0, \qquad P_1, \dots, P_\ell \in \mathbb{Z}[y], \qquad P_i(0) = 0, \qquad 1 \le \deg P_1 < \cdots < \deg P_\ell = d. \tag{1}$$

We write $[N] = \{1, \dots, N\}$, $\mathbb{E}_{n \le N} = N^{-1} \sum_{n \le N}$, and $\mathbf{P} = (P_1, \dots, P_\ell)$. Unrestricted sums over an integer variable are over $\mathbb{Z}$.

Theorem 1.1. *For every family of polynomials satisfying* (1)*, there are $c_* > 1$ and $\kappa > 0$ with the following property. For every $1 < c < c_*$, there exists $N_0 = N_0(\mathbf{P}, c)$ such that, for every integer $N \geq N_0$, every $A \subset [N]$ satisfying*

$$|A| \geq \frac{N}{(\log N)^{\kappa}}$$

contains a configuration

$$x, \quad x + P_1(p-1), \quad \ldots, \quad x + P_\ell(p-1) \tag{2}$$

for some $p \in \mathbb{P}_c$, with the displayed terms pairwise distinct.

The connection between polynomial configurations and multiple recurrence suggests a corresponding pointwise question: do simultaneous polynomial returns have a limiting frequency along almost every orbit when the prime parameter is restricted to a set of power-saving zero relative density?

For ordinary primes, pointwise convergence of single linear averages goes back to Bourgain and Wierdl [4, 24]. Wooley and Ziegler proved norm convergence for multiple polynomial averages along primes [25]. Pointwise convergence was subsequently obtained by Krause, the author, Tao and Teräväinen for the bilinear family $(n, P(n))$, $\deg P \geq 2$ [8], and by Mirek, Wan and Wright for general distinct-degree families [15].

For thin prime sets, Mirek established single pointwise ergodic theorems, including along Piatetski–Shapiro primes [14]. Fornal and Krause treated bilinear linear averages along primes and their Piatetski–Shapiro restrictions [6]. The author proved a distinct-degree pointwise theorem along the Piatetski–Shapiro integers in [16, Theorem 1.2]. Our next theorem shows that, for $c > 1$ sufficiently close to 1, restricting to the polynomially sparser set $\mathbb{P}_c$ preserves both pointwise convergence and the limit of the ordinary-prime averages.

Let (X, ν, T) be an invertible probability measure-preserving system and let $f_1, \ldots, f_\ell \in L^\infty(X)$. Put

$$A_N^{\mathbb{P}_c}(\mathbf{f})(x) := \frac{1}{\pi_c(N)} \sum_{\substack{p \leq N \\ p \in \mathbb{P}_c}} \prod_{i=1}^{\ell} f_i\big(T^{P_i(p-1)}x\big), \qquad \pi_c(N) := |\mathbb{P}_c \cap [N]|. \tag{3}$$

Theorem 1.2 (Pointwise averages along Piatetski–Shapiro primes)**.** *After decreasing $c_* > 1$ if necessary, let $1 < c < c_*$. Then the averages* (3) *converge almost everywhere, and their limit agrees almost everywhere with the limit of*

$$\frac{1}{\pi(N)} \sum_{p \leq N} \prod_{i=1}^{\ell} f_i\big(T^{P_i(p-1)}x\big). \tag{4}$$

Here $\pi(N) = |\mathbb{P} \cap [N]|$.

The comparison underlying this theorem also implies lacunary r-variation bounds for the weighted averages when $r > 2$; see Corollary 4.2.

Remark 1.3. *We expect Theorem 1.2 to extend to $f_i \in L^{p_i}(X)$ with $p_i > 1$ and $\sum_{i=1}^{\ell} p_i^{-1} \leq 1$, allowing c_* to depend on the exponents. Such extensions lie beyond the scope of this paper. The L^1 endpoint is more delicate, even in the single-function case: LaVictoire proved that pointwise convergence can fail for linear averages along*

the ordinary primes [10]. *Mirek raised the corresponding endpoint question for thin prime sets* [14, discussion following Theorem 1.14].

Both arguments begin by approximating the normalized Piatetski–Shapiro-prime weight by the ordinary prime weight. Writing $\gamma = 1/c$ and letting Λ denote the von Mangoldt function, put

$$\Lambda_\gamma(n) := \gamma^{-1} n^{1-\gamma} \Lambda(n) \mathbf{1}_{\mathcal{S}_c}(n), \qquad D_\gamma := \Lambda_\gamma - \Lambda. \tag{5}$$

The normalization compensates for the sparsity of $\mathbb{P}_c$. Centering in this way reduces both problems to showing that the error weight D_γ contributes negligibly to the relevant polynomial averages.

A customary approach, based on PET induction and generalized von Neumann estimates, controls this contribution by a Gowers norm of the error weight. Here it is important to retain a power saving despite the growing normalization factor $n^{1-\gamma}$. We instead exploit the distinct degrees: Teräväinen's theorem controls the error directly through its correlations with ordinary polynomial phases, avoiding the need for a quantitative Gowers-uniformity estimate for D_γ.

Write $\mathrm{e}(t) = e^{2\pi i t}$ and define

$$\|w\|_{u^s[M]} := \sup_{\substack{\phi \in \mathbb{R}[y] \\ \deg \phi < s}} \left| \mathbb{E}_{m \le M} w(m) \mathrm{e}(-\phi(m)) \right|. \tag{6}$$

A function is called 1-bounded if its absolute value is at most 1.

Theorem 1.4 (Teräväinen, [23, Theorem 4.1])**.** *Let $R_1, \dots, R_\ell \in \mathbb{Z}[y]$ satisfy $\deg R_1 < \cdots < \deg R_\ell = d$, and put $\mathbf{R} = (R_1, \dots, R_\ell)$. Let $C_0 \ge 1$. There is an integer $1 \le K \ll_d 1$ such that, whenever $F_0, \dots, F_\ell : \mathbb{Z} \to \mathbb{C}$ are 1-bounded and supported on $[-C_0 M^d, C_0 M^d]$, and $\theta : [M] \to \mathbb{C}$ is 1-bounded,*

$$\left| \frac{1}{M^{d+1}} \sum_{x \in \mathbb{Z}} \sum_{m \le M} \theta(m) F_0(x) \prod_{i=1}^{\ell} F_i(x + R_i(m)) \right| \ll_{C_0, \mathbf{R}} \left(M^{-1} + \|\theta\|_{u^{d+1}[M]} \right)^{1/K}. \tag{7}$$

By homogeneity, the same average with θ replaced by a weight g satisfying $|g| \le B$, $B > 0$, is bounded, up to the same implied constant, by

$$B \left(M^{-1} + B^{-1} \|g\|_{u^{d+1}[M]} \right)^{1/K}. \tag{8}$$

Shao and Sun's estimate [21, Theorem 2.1] supplies a power saving for the correlations of D_γ with polynomial phases, uniformly in their real coefficients.

For Theorem 1.1, we insert this comparison into the density increment argument of [9]. Its PET induction comparison estimates and density increments accommodate the customary (C, Q)-coefficient condition, which records the coefficient growth caused by restricting the original set to arithmetic progressions. In our iteration, the resulting polynomials have the special form

$$P_{i,Q}(y) := \frac{P_i(Qy)}{Q}. \tag{9}$$

Applying Theorem 1.4 directly to this changing family would introduce constants depending on Q. Instead, the change of variables $n = Qm$, $z = Qx$, with the functions supported on multiples of Q, restores the original fixed polynomials P_i. The resulting loss is only a power of Q. Since $Q = N^{o(1)}$ throughout the iteration, Shao–Sun's power saving absorbs this loss. Thus no new coefficient-uniform version

of Theorem 1.4 is needed. The comparison allows the absence of a sparse-prime configuration to force a density increment; repeated increments then give the required contradiction.

For Theorem 1.2, transference gives a power-saving L^1 estimate for the difference between the two weighted ergodic averages. The ordinary-prime averages converge by [15, Theorem 1.1], so the sparse-prime averages have the same limit. Finally, removing higher prime powers and applying partial summation gives the unweighted version of the ergodic theorem.

Section 2 proves the resulting comparison with ordinary prime counts. Section 3 inserts that comparison into the density increment of [9]. Section 4 proves Theorem 1.2.

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2. The Piatetski–Shapiro-prime comparison

Recall the notation of (5) and extend the weights by zero to non-positive integers. Contributions from higher prime powers will be removed by elementary estimates. We first record the arithmetic input.

Theorem 2.1 (Shao–Sun polynomial-phase estimate)**.** *For every $r \geq 1$ there are $\varepsilon_r, \sigma_r > 0$ such that, if $1 - \varepsilon_r < \gamma < 1$, then*

$$\sup_{\deg \phi \leq r} \left| \sum_{n \leq X} D_\gamma(n) \mathrm{e}(\phi(n)) \right| \ll_{r,\gamma} X^{1-\sigma_r}. \tag{10}$$

The case $r = 1$ goes back to Balog–Friedlander [2], while polynomial phases of general degree were studied by Akbal–Güloğlu [1]. We obtain (10) from the Shao–Sun nilsequence estimate [21, Theorem 2.1].

Proof. Take the additive group $G = \mathbb{R}$, the lattice $\Gamma = \mathbb{Z}$, and the filtration

$$G_0 = \cdots = G_r = \mathbb{R}, \qquad G_j = \{0\} \quad (j > r).$$

With the standard basis, this is a filtered nilmanifold of degree r, dimension 1, and complexity $O_r(1)$. The function $F(t + \mathbb{Z}) := \mathrm{e}(t)$ has Lipschitz norm $O(1)$.

Every real polynomial ϕ of degree at most r admits an expansion

$$\phi(n) = \sum_{j=0}^{r} a_j \binom{n}{j}, \qquad a_j \in \mathbb{R}.$$

Thus $g_\phi(n) := \phi(n)$ is a polynomial sequence adapted to this filtration, in the sense of [21, Definition 4.5]. Choose a fixed $\delta_r > 0$, depending only on r, such that the nilmanifold complexity and $\|F\|_{\mathrm{Lip}}$ are at most δ_r^{-1}.

Applying [21, Theorem 2.1] with degree r, dimension 1, and these fixed data gives $\varepsilon_r, \sigma_r > 0$ such that, whenever $1 - \varepsilon_r < \gamma < 1$,

$$\left| \sum_{n \le X} \big(\Lambda_\gamma(n) - \Lambda(n)\big) F(g_\phi(n) + \Gamma) \right| \ll_{r,\gamma} X^{1-\sigma_r}.$$

Here $F(g_\phi(n) + \Gamma) = \mathrm{e}(\phi(n))$. The estimate is uniform in g_ϕ: the cited theorem imposes no coefficient bound on the polynomial sequence, while all other data above depend only on r. Taking the supremum over $\deg \phi \le r$ proves (10). □

The next lemma restricts the estimate to the progressions arising in the density increment.

Lemma 2.2. *For every integer $r \ge 1$, there are $\varepsilon_r, \sigma_r > 0$ such that, whenever $1 - \varepsilon_r < \gamma < 1$ and $Q, M \ge 1$ are integers, one has*

$$\sup_{\substack{\psi \in \mathbb{R}[y] \\ \deg \psi \le r}} \left| \sum_{y \le M} D_\gamma(Qy+1)\mathrm{e}(\psi(y)) \right| \ll_{r,\gamma} (QM)^{1-\sigma_r}. \tag{11}$$

Proof. Write $L \coloneqq QM$. The change of variables $n = Qy + 1$ implies that

$$\sum_{y \le M} D_\gamma(Qy+1)\mathrm{e}(\psi(y)) = \sum_{\substack{2 \le n \le L+1 \\ n \equiv 1 \pmod Q}} D_\gamma(n)\mathrm{e}\left(\psi\left(\frac{n-1}{Q}\right)\right).$$

By orthogonality,

$$\mathbf{1}_{\{n \equiv 1 \pmod Q\}} = \frac{1}{Q} \sum_{a \bmod Q} \mathrm{e}\left(\frac{a(n-1)}{Q}\right).$$

For each a, the phase

$$\psi\left(\frac{n-1}{Q}\right) + \frac{a(n-1)}{Q}$$

is a real polynomial in n of degree at most r. Since $D_\gamma(1) = 0$, we may include $n = 1$ in each inner sum. Theorem 2.1, applied with $X = L + 1$, gives

$$\sup_{a \bmod Q} \left| \sum_{n=1}^{L+1} D_\gamma(n)\mathrm{e}\left(\psi\left(\frac{n-1}{Q}\right) + \frac{a(n-1)}{Q}\right) \right| \ll_{r,\gamma} L^{1-\sigma_r}.$$

Averaging over the Q additive characters and using the uniformity in ψ proves the lemma. □

For the fixed family $\mathbf{P}$ in (1), recall the rescaled polynomials $P_{i,Q}$ from (9). For a weight $w : \mathbb{N} \to \mathbb{C}$ and finitely supported functions $\mathbf{f} = (f_0, \dots, f_\ell)$ on $\mathbb{Z}$, put

$$\mathcal{T}_{N,M,Q}(w; \mathbf{f}) := \frac{1}{N} \sum_{x \in \mathbb{Z}} \mathbb{E}_{y \le M} w(Qy+1) f_0(x) \prod_{i=1}^{\ell} f_i\big(x + P_{i,Q}(y)\big). \tag{12}$$

Proposition 2.3. *Fix the family (1), $C_0 \ge 1$, and $0 < a < 1$. There are constants $c_* = c_*(d) > 1$ and $\xi = \xi(d) > 0$ with the following property. Let $1 < c < c_*$, put $\gamma = 1/c$, and let N, Q be positive integers with $Q \le \exp((\log N)^a)$. Set*

$$M := \left\lfloor \left(\frac{N}{Q^{d-1}}\right)^{1/d} \right\rfloor.$$

If M is sufficiently large in terms of $c, \mathbf{P}, C_0, a$ and $f_0, \dots, f_\ell$ are 1-bounded and supported on $[-C_0N, C_0N]$, then

$$|\mathcal{T}_{N,M,Q}(D_\gamma; \mathbf{f})| \ll_{c,\mathbf{P},C_0} N^{-\xi}. \tag{13}$$

Proof. Put $L = QM$. Since $Q = N^{o(1)}$,

$$L^d \asymp QN, \qquad \frac{L^{d+1}}{NM} \asymp Q^2. \tag{14}$$

For $0 \le i \le \ell$, define

$$F_i(Qx) = f_i(x), \qquad F_i(z) = 0 \quad \text{if } Q \nmid z.$$

These functions are 1-bounded and supported on $[-C_1L^d, C_1L^d]$ for a constant $C_1 = C_1(C_0, \mathbf{P})$. Furthermore,

$$F_i\big(Qx + P_i(Qy)\big) = f_i\big(x + P_{i,Q}(y)\big). \tag{15}$$

For $n \le L$, set

$$\Theta(n) := B^{-1} D_\gamma(n+1) \mathbf{1}_{\{Q|n\}}, \qquad B := C_2 L^{1-\gamma} \log(2L),$$

where $C_2 = C_2(\gamma)$ is chosen so that $|\Theta| \le 1$. By Lemma 2.2,

$$\|\Theta\|_{u^{d+1}[L]} = \frac{1}{BL} \sup_{\deg \phi \le d} \left| \sum_{y \le M} D_\gamma(Qy+1) \mathrm{e}(-\phi(Qy)) \right| \ll_{c,d} B^{-1} L^{-\sigma_d}. \tag{16}$$

Note that every polynomial phase in y occurs as $\phi(Qy)$ for a real polynomial ϕ, so no restriction of the supremum is hidden in this identity. Using (15), (16), and the definitions of F_i and Θ, followed by Theorem 1.4, we obtain

$$\begin{aligned} |\mathcal{T}_{N,M,Q}(D_\gamma; \mathbf{f})| &= \frac{B}{NM} \left| \sum_{z \in \mathbb{Z}} \sum_{n=1}^{L} \Theta(n) F_0(z) \prod_{i=1}^{\ell} F_i(z + P_i(n)) \right| \\ &\ll_{c,\mathbf{P},C_0} B \frac{L^{d+1}}{NM} \left(L^{-1} + B^{-1} L^{-\sigma_d} \right)^{1/K} \\ &\ll_{c,\mathbf{P},C_0} BQ^2 \left(L^{-1} + B^{-1} L^{-\sigma_d} \right)^{1/K}. \end{aligned}$$

To make the final saving explicit, choose γ so close to 1 that

$$1 - \gamma < \frac{\min(1, \sigma_d)}{8K}.$$

For sufficiently large L we then have $B \ll_c L^{2(1-\gamma)}$. Splitting the last parenthesis into its two terms shows that

$$|\mathcal{T}_{N,M,Q}(D_\gamma; \mathbf{f})| \ll_{c,\mathbf{P},C_0} Q^2 \left(L^{2(1-\gamma)-1/K} + L^{2(1-\gamma)(1-1/K)-\sigma_d/K} \right).$$

Both powers of L are negative. Since $Q = L^{o(1)}$ and $L = N^{1/d+o(1)}$, this is $O_{c,\mathbf{P},C_0}(N^{-\xi})$ for some $\xi > 0$. □

3. Quantitative recurrence

We now insert Proposition 2.3 into the distinct-degree density increment of [9]. We state the exact interface that we use.

For an integer $s \geq 2$, write $\mathcal{L}(s) := \operatorname{lcm}(1, \dots, s)$, and let $P^+(Q)$ be the largest prime factor of Q, with $P^+(1) = 1$. For $C \geq 1$ and an integer $Q \geq 1$, an integer polynomial $R(y) = a_e y^e + \cdots + a_1 y$ of degree $e \geq 1$ has (C, Q)-coefficients if there exists $a'_e \in \mathbb{Z}$ such that

$$a_e = a'_e Q^{e-1}, \qquad 0 < |a'_e| \leq C, \qquad |a_j| \leq C|a_e| \quad (1 \leq j < e).$$

For the fixed family (1), choose $C \geq 1$ bounding the absolute values of all coefficients of $P_1, \dots, P_\ell$. Then $P_{i,Q}$ has (C, Q)-coefficients for every integer $Q \geq 1$ and $1 \leq i \leq \ell$. Indeed, writing $d_i = \deg P_i$ and $P_i(y) = \sum_{j=1}^{d_i} a_{i,j} y^j$, we have

$$P_{i,Q}(y) = \sum_{j=1}^{d_i} a_{i,j} Q^{j-1} y^j. \tag{17}$$

The next proposition converts a discrepancy in ordinary-prime counts into a density increment on an arithmetic progression. The proof closely follows [9, Section 5], combining Proposition 5.2 with the density increment argument in the proof of Lemma 5.4.

Proposition 3.1. *Fix C, d and ℓ. There are constants $C_1, K_1 \geq 2$, $0 < c_0 < 1/2$, and $c_1, c_2 > 0$ with the following property. Suppose N is sufficiently large and*

$$(\log N)^{-1/C_1^2} < \delta \leq 1, \qquad \lceil K_1 \delta^{-C_1} \rceil \leq s \leq (\log N)^{c_0},$$

and

$$Q \leq \exp((\log N)^{2c_0}), \qquad P^+(Q) \leq s, \qquad \mathcal{L}(s) \mid Q.$$

Let $M = \lfloor (N/Q^{d-1})^{1/d} \rfloor$, and let $R_0 = 0, R_1, \dots, R_\ell$ have distinct increasing degrees, (C, Q)-coefficients, and largest degree d. If $A \subset [N]$ has $|A| = \delta N$ and

$$\begin{aligned} &\Bigg| \frac{1}{N} \sum_x \mathbb{E}_{y \leq M} \Lambda(Qy+1) \prod_{i=0}^{\ell} \mathbf{1}_A(x + R_i(y)) \\ &\quad - \frac{1}{N} \sum_x \mathbb{E}_{y \leq M} \Lambda(Qy+1) \prod_{i=0}^{\ell} \delta \mathbf{1}_{[N]}(x + R_i(y)) \Bigg| \geq c_1 \delta^{\ell+1} \mathbb{E}_{y \leq M} \Lambda(Qy+1), \end{aligned} \tag{18}$$

then there are integers $b = O_d(1)$, $N' \geq 1$, and a such that

$$a + Q^b[N'] \subset [N], \qquad |A \cap (a + Q^b[N'])| \geq (1 + c_2)\delta N', \tag{19}$$

and, writing $r = \deg R_1$,

$$N' \geq Q^{-O_{d,\ell}(1)} \delta^{O_{d,\ell}(1)} M^r \exp(-d\sqrt{\log M}). \tag{20}$$

Proof. Fix c_1 as in [9, Lemma 5.1], and let c_0 denote the constant c' chosen in [9, (5.7)]. Take C_1, K_1 as in [9, (5.8)], including the enlargements in [9, (5.35)]. By [9, Proposition 5.2, (5.12)], hypothesis (18) gives an integer M_0 satisfying

$$M \exp(-\sqrt{\log M}) \leq M_0 \leq M \tag{21}$$

for which

(22)
$$\left|\frac{1}{N}\sum_{x\in\mathbb{Z}}\mathbb{E}_{y\le M_0}\prod_{i=0}^{\ell}\mathbf{1}_A(x+R_i(y))-\frac{1}{N}\sum_{x\in\mathbb{Z}}\mathbb{E}_{y\le M_0}\prod_{i=0}^{\ell}\delta\mathbf{1}_{[N]}(x+R_i(y))\right|\ge\frac{c_1}{4}\delta^{\ell+1}.$$

Apply [9, Lemma 5.3] with $\varepsilon=(c_1/16)\delta^{3(\ell+1)}$ and all inputs equal to $\mathbf{1}_A$. The choices of C_1,K_1, together with $s\ge\lceil K_1\delta^{-C_1}\rceil$ and $\mathcal{L}(s)\mid Q$, ensure the divisibility condition [9, (5.25)]. The alternative [9, (5.26)] would imply $\log M_0\ll_{C,d,\ell}(\log N)^{2c_0}+\log\log N=o(\log N)$, contrary to $\log M_0\ge(3d)^{-1}\log N$ for sufficiently large N.

Thus we obtain partitions $\mathcal{B}_0,\dots,\mathcal{B}_\ell$ of $[N]$ into arithmetic progressions of common difference Q^b, where $1\le b\ll_d 1$. By [9, (5.28)], every atom has length at least

$$|B|\ge Q^{-O_{d,\ell}(1)}\delta^{O_{d,\ell}(1)}M_0^{\deg R_1}. \tag{23}$$

Let F_i equal the density of A on each atom of $\mathcal{B}_i$, extended by zero outside $[N]$. The approximation [9, (5.27)] and (22) give

$$\left|\frac{1}{N}\sum_{x}\mathbb{E}_{y\le M_0}\left(\prod_{i=0}^{\ell}F_i(x+R_i(y))-\prod_{i=0}^{\ell}\delta\mathbf{1}_{[N]}(x+R_i(y))\right)\right|\ge\frac{c_1}{8}\delta^{\ell+1}. \tag{24}$$

We argue as in [9, proof of Lemma 5.4, immediately after (5.43)]. Fix $c_2>0$ so that $2(\ell+1)c_2(1+c_2)^\ell<c_1/8$, and suppose that $F_i\le(1+c_2)\delta$ for every i. Since $\sum_x F_i(x)=\delta N$, the positive and negative parts of $F_i-\delta\mathbf{1}_{[N]}$ have equal mass, giving

$$\sum_x|F_i(x)-\delta\mathbf{1}_{[N]}(x)|\le 2c_2\delta N.$$

Expanding the difference of products one factor at a time bounds the left-hand side of (24) by

$$\frac{((1+c_2)\delta)^\ell}{N}\sum_{i=0}^{\ell}\sum_x|F_i(x)-\delta\mathbf{1}_{[N]}(x)|\le 2(\ell+1)c_2(1+c_2)^\ell\delta^{\ell+1}.$$

The resulting contradiction shows that some F_i exceeds $(1+c_2)\delta$ on an atom $a+Q^b[N']\subset[N]$. By the definition of F_i, this gives (19).

Writing $r=\deg R_1$, the resolution bound from (23) gives $N'\ge Q^{-O_{d,\ell}(1)}\delta^{O_{d,\ell}(1)}M_0^r$. Combining this with the lower bound for M_0 in (21) and $r\le d$ proves (20). □

The next lemma verifies the discrepancy hypothesis using the absence of a Piatetski–Shapiro-prime configuration.

Lemma 3.2. *Fix the family* (1). *Choose $C\ge1$ bounding the absolute values of all coefficients of $P_1,\dots,P_\ell$, and fix the corresponding constants C_1,K_1,c_0,c_1 from Proposition 3.1. This choice of C gives a common coefficient bound for every rescaled family* (9). *There exists $c_*=c_*(\deg P_\ell)>1$ such that the following holds for every $1<c<c_*$, with $\gamma=1/c$. Let N be a sufficiently large positive integer in terms of $c,\mathbf{P}$, and let $A\subset[N]$ have density $\delta=|A|/N$ satisfying $(\log N)^{-1/C_1^2}<\delta\le1$. Let $s\ge2$ be an integer satisfying*

$$\lceil K_1\delta^{-C_1}\rceil\le s\le(\log N)^{c_0},$$

and let $Q\ge1$ be an integer satisfying

$$Q\le\exp((\log N)^{2c_0}),\qquad P^+(Q)\le s,\qquad \mathcal{L}(s)\mid Q.$$

Put $M = \lfloor (N/Q^{d-1})^{1/d} \rfloor$. If A contains no configuration of pairwise distinct elements

$$x, \quad x + P_{1,Q}(y), \quad \dots, \quad x + P_{\ell,Q}(y) \tag{25}$$

with $x \in \mathbb{Z}$, $y \in \mathbb{N}$, and $Qy + 1 \in \mathbb{P}_c$, then (18) holds with $R_0 = 0$ and $R_i = P_{i,Q}$ for $1 \le i \le \ell$.

Proof. First we establish a lower bound for the ordinary-prime reference count, with all inputs equal to $\delta \mathbf{1}_{[N]}$.

Choose a sufficiently small fixed $\tau = \tau(\mathbf{P}) > 0$ and put $M_\tau = \lfloor \tau M \rfloor$. The coefficient bounds and the definition of M give

$$\max_{1 \le i \le \ell} |P_{i,Q}(y)| \le N/10 \qquad (1 \le y \le M_\tau).$$

Hence, for each such y, at least $N/2$ integers x satisfy $x + P_{i,Q}(y) \in [N]$ for every $0 \le i \le \ell$. Therefore

$$\begin{aligned} &\frac{1}{N} \sum_x \mathbb{E}_{y \le M} \Lambda(Qy+1) \prod_{i=0}^{\ell} \delta \mathbf{1}_{[N]}(x + P_{i,Q}(y)) \\ &\qquad \ge \frac{\delta^{\ell+1}}{2M} \sum_{y \le M_\tau} \Lambda(Qy+1) \gg_{\mathbf{P}} \delta^{\ell+1} \mathbb{E}_{y \le M} \Lambda(Qy+1). \end{aligned} \tag{26}$$

The last inequality is the ordinary-prime mass comparison [9, (5.2)].

Taking $\mathbf{f} = (\mathbf{1}_A, \dots, \mathbf{1}_A)$, the identity $D_\gamma = \Lambda_\gamma - \Lambda$ and Proposition 2.3 give

$$\mathcal{T}_{N,M,Q}(\Lambda; \mathbf{f}) = \mathcal{T}_{N,M,Q}(\Lambda_\gamma; \mathbf{f}) + O_{c,\mathbf{P}}(N^{-\xi}). \tag{27}$$

Since Λ_γ is supported on $\mathcal{S}_c$, every nonzero contribution on the right with $Qy + 1$ prime comes from $\mathbb{P}_c$. The hypothesis (25) eliminates these terms whenever $0, P_{1,Q}(y), \dots, P_{\ell,Q}(y)$ are pairwise distinct. So only collisions and higher prime powers remain.

Each polynomial $P_{i,Q}$, $1 \le i \le \ell$, and each difference $P_{i,Q} - P_{j,Q}$, $1 \le i < j \le \ell$, is nonzero and has degree at most d. Hence collisions occur for only $O_{d,\ell}(1)$ values of y. Their total contribution is

$$\ll_{c,\mathbf{P}} \frac{(QM)^{1-\gamma} \log(2QM)}{M}. \tag{28}$$

The contribution of higher prime powers is bounded, using the elementary estimate in [9, (5.5)] and $\Lambda_\gamma(n) \ll_\gamma (QM)^{1-\gamma} \Lambda(n)$ for $n \le QM + 1$, by

$$\ll_\gamma Q^{3/2-\gamma} M^{1/2-\gamma} (\log(2QM))^2. \tag{29}$$

Finally, the prime-mass estimate[1] [9, (5.15)] gives

$$\mathbb{E}_{y \le M} \Lambda(Qy+1) \gg \exp\bigl(-O((\log N)^{2c_0})\bigr). \tag{30}$$

Thus $N^{-\eta}$ is negligible compared with the right-hand side of (26), uniformly in the density range of Proposition 3.1. The constant c_1 in [9, (5.11)] is chosen as a fixed sufficiently small fraction of the implicit constant in (26). The preceding estimates

[1]The recent exclusion of Landau–Siegel zeros, originating in an OpenAI argument and presented by Green [7] (see also Tao's exposition [22]), combined with the prime number theorem in arithmetic progressions, gives the stronger bound $\mathbb{E}_{y \le M} \Lambda(Qy+1) \gg Q/\varphi(Q)$ in the present parameter range. For the sake of completeness, we retain the classical bounds in the proof.

show that $\mathcal{T}_{N,M,Q}(\Lambda;\mathbf{f}) = O_{c,\mathbf{P}}(N^{-\eta})$. Subtracting this count in A from the count in (26), we obtain, for sufficiently large N,

$$\begin{aligned}\frac{1}{N}\sum_x \mathbb{E}_{y\le M}\Lambda(Qy+1)\prod_{i=0}^{\ell}\delta\mathbf{1}_{[N]}(x+P_{i,Q}(y))\\ -\frac{1}{N}\sum_x \mathbb{E}_{y\le M}\Lambda(Qy+1)\prod_{i=0}^{\ell}\mathbf{1}_{A}(x+P_{i,Q}(y))\\ \ge c_1\delta^{\ell+1}\mathbb{E}_{y\le M}\Lambda(Qy+1).\end{aligned}$$

This proves (18). □

Proof of Theorem 1.1. Let $\alpha = |A|/N$ and suppose, towards a contradiction, that A contains no configuration (2) with pairwise distinct terms. Let C_1, K_1, c_0, c_2 be supplied by Proposition 3.1, and put

$$s = \left\lceil K_1\alpha^{-C_1}\right\rceil, \qquad Q_0 = \mathcal{L}(s).$$

Since $Q_0 = \mathcal{L}(s) \le s!$, we have

$$\log Q_0 \le s\log s \ll_{\mathbf{P}} \alpha^{-O_{\mathbf{P}}(1)}.$$

The assumption $\alpha \ge (\log N)^{-\kappa}$ therefore gives $\log Q_0 = o(\log N)$, provided $\kappa > 0$ is sufficiently small. Thus $Q_0 = N^{o(1)}$.

Choose a residue class modulo Q_0 on whose intersection with $[N]$ the set A has maximal relative density, and represent it by $r_0 \in \{1-Q_0,\dots,0\}$. Put

$$N_0 := \left\lfloor \frac{N-r_0}{Q_0}\right\rfloor, \qquad A_0 := \{u \in [N_0] : r_0 + Q_0 u \in A\}.$$

Then $r_0 + Q_0[N_0] = (r_0 + Q_0\mathbb{Z}) \cap [N]$. Averaging over the full intersections of the residue classes with $[N]$ gives $\delta_0 := |A_0|/N_0 \ge \alpha$. Moreover, for sufficiently large N,

$$N_0 \ge N/Q_0 - 1 \ge N/(2Q_0). \tag{31}$$

Since $r_0 + Q_0A_0 \subseteq A$, the forbidden-configuration hypothesis (2) on A implies that A_0 contains no configuration of pairwise distinct elements

$$x,\ x+P_{1,Q_0}(y),\dots,x+P_{\ell,Q_0}(y), \qquad y\in\mathbb{N},\quad Q_0y+1\in\mathbb{P}_c.$$

Indeed, if such a configuration existed, setting $a = r_0 + Q_0x$ and $p = Q_0y+1$ would give $a \in A$ and

$$a + P_i(p-1) = r_0 + Q_0\big(x + P_{i,Q_0}(y)\big) \in A \qquad (1\le i\le \ell).$$

Multiplication by Q_0 preserves distinctness, so these elements would form a configuration (2) in A, a contradiction.

We iterate Lemma 3.2 and Proposition 3.1. Suppose that $A_j \subset [N_j]$ has density $\delta_j \ge \alpha$ and contains no configuration of pairwise distinct elements

$$x,\ x+P_{1,Q_j}(y),\dots,x+P_{\ell,Q_j}(y), \qquad y\in\mathbb{N},\quad Q_jy+1\in\mathbb{P}_c.$$

The density increment gives a progression $a_j + Q_j^{b_j}[N_{j+1}] \subset [N_j]$, with $b_j = O_d(1)$, on which A_j has density at least $(1+c_2)\delta_j$. Put

$$A_{j+1} := \{u \in [N_{j+1}] : a_j + Q_j^{b_j}u \in A_j\}, \qquad \delta_{j+1} := \frac{|A_{j+1}|}{N_{j+1}},$$

and set

$$Q_{j+1} := Q_j^{b_j+1}. \tag{32}$$

In particular, the first step produces A_1, and $\delta_{j+1} \geq (1+c_2)\delta_j$ at every step.

The forbidden-configuration (2) hypothesis is preserved, since

$$Q_j^{b_j} P_{i,Q_{j+1}}(y) = P_{i,Q_j}(Q_j^{b_j} y), \qquad Q_{j+1}y + 1 = Q_j(Q_j^{b_j} y) + 1.$$

Thus multiplication by $Q_j^{b_j}$ and translation by a_j would take a forbidden configuration in A_{j+1} to one in A_j.

We keep the smoothness parameter s fixed throughout. This is permitted by Proposition 3.1, because

$$\lceil K_1 \delta_j^{-C_1} \rceil \leq \lceil K_1 \alpha^{-C_1} \rceil = s.$$

Moreover, every Q_j is a power of Q_0, so

$$P^+(Q_j) = P^+(Q_0) \leq s \ll_{\mathbf{P}} \alpha^{-C_1}, \qquad \mathcal{L}(s) \mid Q_j.$$

Hence the smoothness bound remains in terms of the initial density α, even as δ_j increases.

We now verify inductively that the hypotheses required for each density increment remain valid. Since $b_j = O_d(1)$ and there are at most $O_{\mathbf{P}}(\log \alpha^{-1} + 1)$ possible increments, [9, (5.57)] gives

$$\log Q_j \leq \alpha^{-O_{\mathbf{P}}(1)}. \tag{33}$$

Writing $r = \deg P_1$ and $\vartheta = r/d$, the length estimate (20) gives

$$\log N_{j+1} \geq \vartheta \log N_j - O_{\mathbf{P}}\big(\log Q_j + \log(\alpha^{-1}) + \sqrt{\log N_j}\big), \tag{34}$$

which is [9, (5.58)]. Iterating this inequality, using (31), (33), and $j = O_{\mathbf{P}}(\log \alpha^{-1}+1)$, gives

$$\log N_j \gg_{\mathbf{P}} \alpha^{O_{\mathbf{P}}(1)} \log N - \alpha^{-O_{\mathbf{P}}(1)} - O_{\mathbf{P}}\big(\log(\alpha^{-1})\sqrt{\log N}\big); \tag{35}$$

see [9, (5.59)]. If $\alpha \geq (\log N)^{-\kappa}$ and $\kappa > 0$ is sufficiently small, the first term in (35) dominates the errors. Hence $\log N_j \geq (\log N)^{\eta_0}$ for some $\eta_0 > 0$, uniformly at every stage. Equations (33) and (35) then imply

$$Q_j \leq \exp((\log N_j)^{2c_0}), \qquad s \leq (\log N_j)^{c_0}, \qquad \delta_j > (\log N_j)^{-1/C_1^2},$$

after decreasing κ and c_0 if necessary. Since every Q_j is a power of Q_0, one also has $P^+(Q_j) \leq s$ and $\mathcal{L}(s) \mid Q_j$. These are exactly the hypotheses of Proposition 3.1. Applying Proposition 2.3 with any fixed a satisfying $2c_0 < a < 1$ also puts every Q_j in its required comparison range. We may therefore perform

$$\left\lfloor \frac{\log(\alpha^{-1})}{\log(1+c_2)} \right\rfloor + 1$$

increments. The resulting density is greater than 1, a contradiction. This proves the theorem. □

4. Pointwise ergodic averages

By homogeneity, assume throughout this section that $\|f_i\|_\infty \le 1$. Write $z_n(x) := \prod_{i=1}^{\ell} f_i(T^{P_i(n-1)}x)$ and put

$$B_N^\gamma(x) := \mathbb{E}_{n\le N}\Lambda_\gamma(n)z_n(x), \qquad B_N(x) := \mathbb{E}_{n\le N}\Lambda(n)z_n(x). \tag{36}$$

Taking the zero phase in Theorem 2.1 and using the prime number theorem gives

$$\mathbb{E}_{n\le N}\Lambda_\gamma(n) = \mathbb{E}_{n\le N}\Lambda(n) + O_{c,d}(N^{-\sigma_d}) = 1 + o(1). \tag{37}$$

Proposition 4.1 (Pointwise comparison)**.** *After decreasing $c_* > 1$ if necessary, for every $1 < c < c_*$ one has $B_N^\gamma - B_N \to 0$ almost everywhere.*

Proof. By definition, $B_N^\gamma - B_N = \mathbb{E}_{n\le N}D_\gamma(n)z_n$. We apply [16, Proposition 1.4] with $g = D_\gamma$ and the polynomials $n \mapsto P_i(n-1)$, which retain their distinct degrees. Thus, for any $B > 0$ satisfying $|D_\gamma(n)| \le B$ on $[N]$,

$$\|B_N^\gamma - B_N\|_{L^1(X)} \ll_{\mathbf{P}} B\left(N^{-1} + B^{-1}\sup_{\deg\phi\le d}|\mathbb{E}_{n\le N}D_\gamma(n)\mathrm{e}(\phi(n))|\right)^{1/K}.$$

We now take $B = C_c N^{1-\gamma}\log(2N)$, with C_c sufficiently large. Theorem 2.1 gives

$$\sup_{\deg\phi\le d}|\mathbb{E}_{n\le N}D_\gamma(n)\mathrm{e}(\phi(n))| \ll_{c,d} N^{-\sigma_d}.$$

Consequently,

$$\begin{aligned}\|B_N^\gamma - B_N\|_{L^1(X)} &\ll_{c,\mathbf{P}} N^{1-\gamma}\log(2N)\left(N^{-1} + N^{\gamma-1-\sigma_d}\right)^{1/K}\\ &\ll_{c,\mathbf{P}} \log(2N)\left(N^{1-\gamma-1/K} + N^{(1-\gamma)(1-1/K)-\sigma_d/K}\right) \ll_{c,\mathbf{P}} N^{-\eta}.\end{aligned} \tag{38}$$

Here, after decreasing c_* so that $1-\gamma \le \min\{1,\sigma_d\}/(2K)$, we may take $\eta = \min\{1,\sigma_d\}/(4K)$.

Choose $A > \max\{1,\eta^{-1}\}$ and put $N_j = \lfloor j^A\rfloor$. By (38),

$$\sum_j \|B_{N_j}^\gamma - B_{N_j}\|_1 < \infty.$$

Tonelli's theorem therefore gives $B_{N_j}^\gamma(x) - B_{N_j}(x) \to 0$ for almost every x.

To pass to all N, let w be either Λ_γ or Λ. Since $|z_n| \le 1$, for $N_j \le N < N_{j+1}$ the corresponding weighted average differs from its value at N_j by at most

$$\frac{N_{j+1}-N_j}{N_j^2}\sum_{n\le N_j} w(n) + \frac{1}{N_j}\sum_{N_j<n\le N_{j+1}} w(n) = o(1),$$

uniformly in x. Here we used (37) and $N_{j+1}/N_j \to 1$. Applying this to both weighted averages proves the full-sequence comparison. □

Proof of Theorem 1.2. By [15, Theorem 1.1(ii)], applied to the polynomials $n \mapsto P_i(n-1)$, the averages B_N converge almost everywhere. Proposition 4.1 gives the same limit, denoted by $L(x)$, for B_N^γ.

We remove the weights by summation by parts. First, higher prime powers contribute at most

$$\sum_{\substack{p^k\le N\\ k\ge 2}}\left(\Lambda_\gamma(p^k) + \Lambda(p^k)\right) \ll_\gamma N^{3/2-\gamma}(\log(2N))^2 = o(N), \tag{39}$$

since $\gamma > 1/2$ after decreasing c_* if necessary.

Fix x for which $B_N^\gamma(x) \to L(x)$, and put $a(t) = \gamma^{-1}t^{1-\gamma}\log t$ for $t \geq 2$. For a prime p, we have $\Lambda_\gamma(p) = a(p)\mathbf{1}_{\mathbb{P}_c}(p)$. Thus removing higher prime powers leaves precisely the sums over $\mathbb{P}_c$.

Define $W(t) := \sum_{p\leq t,\ p\in\mathbb{P}_c} a(p)$. Estimate (39), together with $|z_n(x)| \leq 1$, gives

$$W(N) = \sum_{n\leq N} \Lambda_\gamma(n) + o(N) = N + o(N),$$
$$\sum_{\substack{p\leq N\\ p\in\mathbb{P}_c}} a(p)z_p(x) = NB_N^\gamma(x) + o(N) = NL(x) + o_x(N).$$

The first asymptotic uses (37); the second uses the convergence of $B_N^\gamma(x)$. Subtracting $L(x)W(N)$ from the second sum therefore yields

$$\sum_{\substack{p\leq N\\ p\in\mathbb{P}_c}} a(p)\big(z_p(x) - L(x)\big) = NL(x) + o_x(N) - L(x)\big(N + o(N)\big) = o_x(N).$$

Since $W(N) \sim N$, the centered weighted sum is also $o_x(W(N))$. Write

$$S(t) := \sum_{\substack{p\leq t\\ p\in\mathbb{P}_c}} a(p)\big(z_p(x) - L(x)\big).$$

For real $t \geq 2$, the conditions $p \leq t$ and $p \leq \lfloor t\rfloor$ are equivalent. Thus $S(t) = S(\lfloor t\rfloor)$ and $W(t) = W(\lfloor t\rfloor)$. Consequently, given $\varepsilon > 0$, there is an integer $t_0 \geq 2$ such that $|S(t)| \leq \varepsilon W(t)$ for every real $t \geq t_0$. For $N \geq t_0$, summation by parts and $a'(t) \geq 0$ give

$$\begin{aligned}
\left|\sum_{\substack{p\leq N\\ p\in\mathbb{P}_c}} \big(z_p(x) - L(x)\big)\right| &= \left|\frac{S(N)}{a(N)} + \int_2^N S(t)\frac{a'(t)}{a(t)^2}\,dt\right| \\
&\leq C_{x,\varepsilon} + \varepsilon\left(\frac{W(N)}{a(N)} + \int_2^N W(t)\frac{a'(t)}{a(t)^2}\,dt\right) \\
&= C_{x,\varepsilon} + \varepsilon\sum_{\substack{p\leq N\\ p\in\mathbb{P}_c}} \frac{a(p)}{a(p)} \\
&= C_{x,\varepsilon} + \varepsilon\pi_c(N).
\end{aligned}$$

Here $C_{x,\varepsilon} := \int_2^{t_0} |S(t)|a'(t)/a(t)^2\,dt$ bounds the initial part of the integral. The equality involving W follows by applying the same summation-by-parts formula to its defining sum.

Since $W(N) \sim N$, the set $\mathbb{P}_c$ is infinite and $\pi_c(N) \to \infty$. Dividing by $\pi_c(N)$ and then letting $\varepsilon \to 0$ proves $A_N^{\mathbb{P}_c}(\mathbf{f})(x) \to L(x)$.

Applying the same summation-by-parts calculation over all primes, with $a(t) = \log t$ and $W(t) = \sum_{p\leq t} \log p \sim t$, shows that the ordinary-prime average (4) also converges to $L(x)$. This proves the theorem. □

For $1 \le r < \infty$ and a sequence $(a_j)_{j\ge 1}$ of complex numbers, define its r-variation seminorm by

$$V^r(a_j : j \ge 1) := \sup_{\substack{J\ge 1 \\ 1\le j_0<\cdots<j_J}} \left(\sum_{h=1}^{J} |a_{j_h} - a_{j_{h-1}}|^r \right)^{1/r}.$$

For a sequence of functions, this definition is applied pointwise.

Corollary 4.2. *Under the hypotheses of Theorem 1.2, let $r > 2$ and $\lambda > 1$. If $(N_j)_{j\ge 1}$ is a sequence of positive integers satisfying $N_{j+1} \ge \lambda N_j$, then*

$$\left\| V^r\big(B^{\gamma}_{N_j} : j \ge 1\big) \right\|_{L^1(X)} \ll_{c,\mathbf{P},r,\lambda} \prod_{i=1}^{\ell} \|f_i\|_\infty.$$

Proof. By homogeneity, assume that $\|f_i\|_\infty \le 1$. The elementary bound $V^r(a_j : j \ge 1) \le 2\sum_j |a_j|$, together with (38), gives

$$\left\| V^r\big(B^{\gamma}_{N_j} - B_{N_j} : j \ge 1\big) \right\|_1 \le 2 \sum_{j\ge 1} \|B^{\gamma}_{N_j} - B_{N_j}\|_1 \ll_{c,\mathbf{P}} \sum_{j\ge 1} N_j^{-\eta} \ll_{c,\mathbf{P},\lambda} 1.$$

Since $\nu(X) = 1$, [15, Theorem 1.1(iv)] and Hölder's inequality give

$$\|V^r(B_{N_j} : j \ge 1)\|_1 \ll_{\mathbf{P},r,\lambda} 1.$$

The triangle inequality completes the proof. □

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